

Practical No. 8

A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user ie. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames 3).

Code:

```
/c/Users/john/OneDrive/Desktop/OSY_CD25D005
GNU nano 8.7
#include <stdio.h>

int main() {
    int pages[50], frames[10];
    int n, framesCount = 3;
    int i, j, k, pos, faults = 0, hits = 0;
    int flag1, flag2;

    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string:\n");
    for(i = 0; i < n; i++) {
        scanf("%d", &pages[i]);
    }

    for(i = 0; i < framesCount; i++)
        frames[i] = -1;

    for(i = 0; i < n; i++) {
        flag1 = flag2 = 0;

        for(j = 0; j < framesCount; j++) {
            if(frames[j] == pages[i]) {
                hits++;
                flag1 = flag2 = 1;
                break;
            }
        }

        if(flag1 == 0) {
            for(j = 0; j < framesCount; j++) {
                if(frames[j] == -1) {
                    faults++;
                    frames[j] = pages[i];
                    flag2 = 1;
                    break;
                }
            }
        }

        if(flag2 == 0) {
            int least[10];
            for(j = 0; j < framesCount; j++) {
                least[j] = -1;
                for(k = i - 1; k >= 0; k--) {
                    if(frames[j] == pages[k]) {
                        least[j] = k;
                        break;
                    }
                }
            }
        }
    }
}
```

```

/c/Users/john/OneDrive/Desktop/OSY_CD25D005
GNU nano 8.7
        flag1 = flag2 = 1;
        break;
    }
}
if(flag1 == 0) {
    for(j = 0; j < framesCount; j++) {
        if(frames[j] == -1) {
            faults++;
            frames[j] = pages[i];
            flag2 = 1;
            break;
        }
    }
}
if(flag2 == 0) {
    int least[10];
    for(j = 0; j < framesCount; j++) {
        least[j] = -1;
        for(k = i - 1; k >= 0; k--) {
            if(frames[j] == pages[k]) {
                least[j] = k;
                break;
            }
        }
    }
    pos = 0;
    for(j = 1; j < framesCount; j++)
        if(least[j] < least[pos])
            pos = j;

    frames[pos] = pages[i];
    faults++;
}

printf("Frame status: ");
for(j = 0; j < framesCount; j++)
    printf("%d ", frames[j]);
printf("\n");
}

float hitPercent = ((float)hits / n) * 100;
float faultPercent = ((float)faults / n) * 100;

printf("\nTotal Page Hits = %d\n", hits);
printf("Total Page Faults = %d\n", faults);
printf("Hit Percentage = %.2f%%\n", hitPercent);
printf("Fault Percentage = %.2f%%\n", faultPercent);

return 0;
}

```

```

/c/Users/john/OneDrive/Desktop/OSY_CD25D005
john@DESKTOP-JAOTCLF MSYS ~
$ C:\Users\john\OneDrive\Desktop\OSY_CD25D005
-bash: C:\Users\john\OneDrive\Desktop\OSY_CD25D005: command not found

john@DESKTOP-JAOTCLF MSYS ~
$ cd C:\Users\john\OneDrive\Desktop\OSY_CD25D005

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ nano lru.c

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ gcc lru.c -o lru

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ ./lru
Enter number of pages: 12
Enter page reference string:
1 2 3 4 1 2 5 1 2 3 4 5
Frame status: 1 -1 -1
Frame status: 1 2 -1
Frame status: 1 2 3
Frame status: 4 2 3
Frame status: 4 1 3
Frame status: 4 1 2
Frame status: 5 1 2
Frame status: 5 1 2
Frame status: 5 1 2
Frame status: 3 1 2
Frame status: 3 4 2
Frame status: 3 4 5

Total Page Hits = 2
Total Page Faults = 10
Hit Percentage = 16.67%
Fault Percentage = 83.33%

john@DESKTOP-JAOTCLF MSYS /c/Users/john/OneDrive/Desktop/OSY_CD25D005
$ |

```